Slips:

1. **Capture error:**

frequently performed, highly habituated sequence of actions accidentally overrides and takes control of a **less frequent, intended sequence of actions** that shares the exact same initial steps.(What you have done thousands times before and became habit)

Ex1: Wrote i++ instead of — (int i=10;i>0;i--).

Ex2: git checkout -b feature-branch to create a new branch. Because their daily habit is checking status, they type git status

2. **Description error:**

a user formulates the correct intention, but performs the action on an incorrect target that looks or feels physically similar to the intended target. (The wrong target looks almost identical to right target)

Ex1: While driving a rental car at night, a driver wants to turn up the radio volume. Because the volume knob and the fan speed knob are the exact same size, shape, and texture, they accidentally turn up the air conditioning instead

Ex2: press delete button because its color is same as save button

3. **Data driven error:**

What you are seeing or hearing at the exact miliseconds**. External information** accidentally enters your action

Ex1: A data analyst is entering financial figures into Excel while listening to a podcast. The host says the number *"fourteen"*. Even though the analyst is looking at a paper invoice that clearly reads *"fifty-six"*, they unconsciously type 14 into the cell.

Ex2: You are typing an important email to a client named Sarah. A Slack notification pops up from a coworker named Dave. You glance at the notification, turn back to your email, and unconsciously type, *"Hi Dave,"* instead of *"Hi Sarah."*

4. **Associative activation error**

when a thought or association in your mind triggers the **wrong action or response**. **Internal thoughts** or memories.

Ex1: You have been thinking about a function that calculates **salary**. You then write a function to calculate **student grades**, but accidentally type:

```
return salary; // Intended: return grade;
```

Ex2: You are writing an email to your **teacher** while thinking about your **project teammate**. You accidentally type your teammate's name in the greeting.

5. **Loss of activation error:**

forget what you intended to do after a distraction

Ex1: You open a new tab to search for something, but a notification distracts you. Then you think: **"What was I going to search for?"**

Ex2: You open a folder to find an assignment, but forget which file you wanted.

6. **Mode errors:**

thinking the system is in one mode, but it is actually in another. The right action for the mode you expected, but the wrong mode is active.

Ex1: You type your password expecting lowercase letters, but **Caps Lock is on**, so uppercase letters are entered.

Ex2: You drag the mouse expecting to draw, but the **eraser tool is selected**, so you erase part of the picture.

Constraints:

1. **Physical:**

shape, size, or structure physically restricts an action. "I cannot put it there because it does not fit.

- **USB-A connector**

Its shape allows it to fit into the port in only one orientation. You cannot normally insert it upside down.

- **RAM installation**

A notch in the RAM module must align with the slot, preventing incorrect orientation.

- **Laptop hinge**

The hinge physically limits how far you can open the screen.

2. **Semantic:**

the meaning of the situation guides your action. It relies on meaning and context. "I should not put it there because its purpose would not make sense." "This action makes sense because of what these things mean."

Imagine a **video-call app** asking you to select a device to capture your face.

You have:

- A **webcam**
- A **microphone**

You choose the **webcam** because it captures images. A microphone captures sound, so it cannot serve that purpose.

3. Cultural:

learned conventions shared by a group or society**.**

Red error message: Signals a problem or warning.

Trash icon: delete

X" on a window: Usually closes the window.

4. Logical:

reasoning and elimination

Forcing function:

1. **Interlock:**

Do A before B is allowed. One action is possible only when a required condition is met.

Ex1: A banking app enables **Confirm Transfer** only after successful OTP verification.

Ex2: Submit stays disabled until an upload finish.

2. **Lock-in:**

Keeps an operation active by preventing you from leaving or stopping it midway. "You started it, finish before leaving."

Ex1: An editor pauses closing when there are unsaved changes, requiring you to choose **Save, Discard, or Cancel** first

Ex2: A payment kiosk prevents returning **Home** until payment processing succeeds or fails.

3. **Lock-out:**

Prevents you from entering a restricted state or performing an action. "You cannot enter or perform this action."

Ex1: An account temporarily blocks login after too many **incorrect password attempts**

Ex2: A **read-only file** cannot be edited.

Ex3: An employee account cannot open the **Admin Panel**.

Interlock = **meet the condition** · Lock-in = **cannot leave yet** · Lockout = **access blocked**.